

AWS Networking Infrastructure

To ensure high availability, the architecture uses a /16 Custom VPC carved into 6 subnets across 2 distinct Availability Zones (AZs), with each subnet provisioned with a /24 block (providing 251 usable IPs per subnet)

VPC CIDR Block: 192.168.0.0/16

- **Subnet Distribution:**
 - **Public Web Tier:** 192.168.1.0/24 (AZ-1) & 192.168.2.0/24 (AZ-2)
 - **Private App Tier:** 192.168.3.0/24 (AZ-1) & 192.168.4.0/24 (AZ-2)
 - **Private DB Tier:** 192.168.5.0/24 (AZ-1) & 192.168.6.0/24 (AZ-2)

Internet Gateway (IGW): Attached to the VPC to enable public routing.

NAT Gateway: Allocated a static Elastic IP (EIP) and deployed strictly inside a Public Subnet (192.168.1.0/24) to give private tiers secure, outbound-only internet access for running package update.

VPC Snap

vpc-081c241c4e4b95465 / MY_PROJECT_VPC

Actions

Details Info			
VPC ID vpc-081c241c4e4b95465	State Available	Block Public Access Off	DNS hostnames Disabled
DNS resolution Enabled	Tenancy default	DHCP option set dopt-063d6cd02c88b3fc8	Main route table rtb-0cec4c7a93ac2fa3b
Main network ACL acl-0913dc52a8129e54c	Default VPC No	IPv4 CIDR 192.168.0.0/16	IPv6 pool -
IPv6 CIDR (Network border group) -	Network Address Usage metrics Disabled	Route 53 Resolver DNS Firewall rule groups -	Owner ID 041944951715
Encryption control ID -	Encryption control mode -		

Subnet Snap

Subnets (6) Info							Last updated 13 minutes ago	Actions	Create subnet
Find subnets by attribute or tag									
	Name	Subnet ID	State	VPC	Block Public				
☐	MY_SUBNET_Privet_DB_1a	subnet-03be8c5ef29e866fe	Available	vpc-081c241c4e4b95465 MY...	Off				
☐	MY_SUBNET_PUBLIC_1a	subnet-06d67f47275c42f33	Available	vpc-081c241c4e4b95465 MY...	Off				
☐	MY_SUBNET_Privet_app_1a	subnet-0330be35645c5a874	Available	vpc-081c241c4e4b95465 MY...	Off				
☐	MY_SUBNET_Privet_DB_1b	subnet-07935d787c185e728	Available	vpc-081c241c4e4b95465 MY...	Off				
☐	MY_SUBNET_Privet_app_1b	subnet-0acf269ff6be0aa05	Available	vpc-081c241c4e4b95465 MY...	Off				
☐	MY_SUBNET_PUBLIC_1b	subnet-04b908b82f753e561	Available	vpc-081c241c4e4b95465 MY...	Off				

EIP Snap

100.56.173.78

Actions

Associate Elastic IP address

Summary

Allocated IPv4 address

100.56.173.78

Type

Public IP

Allocation ID

eipalloc-07404baad031f411e

Reverse DNS record

–

Association ID

eipassoc-0f69d7d2a7b844d2e

Scope

VPC

Associated instance ID

–

Private IP address

192.168.1.43

Network interface ID

eni-0a5ee791e96c333ff

Network interface owner account ID

041944951715

Public DNS

–

NAT Gateway ID

nat-06a029bfa4b4d33e (MY_N-GTW)

Address pool

Amazon

Network border group

us-east-1

Service managed

–

Internet Gateway Snap

igw-0b07763435d1fb017 / MY_IGW

Actions

Details

Internet gateway ID

igw-0b07763435d1fb017

State

Attached

VPC ID

vpc-081c241c4e4b95465 / MY_PROJECT_VPC

Owner

041944951715

Tags (1)

Search tags

Manage tags

< 1 > ⚙

Key

Value

Name

MY_IGW

NAT Gateway Snap

nat-06a029bfa4b4d33e / MY_N-GTW

Actions

Details

NAT gateway ID

nat-06a029bfa4b4d33e

Connectivity type

Public

State

Available

State message

–

NAT gateway ARN

arn:aws:ec2:us-east-1:041944951715:natgateway/nat-06a029bfa4b4d33e

Primary public IPv4 address

100.56.173.78

Primary private IPv4 address

192.168.1.43

Primary network interface ID

eni-0a5ee791e96c333ff

VPC

vpc-081c241c4e4b95465 / MY_PROJECT_VPC

Subnet

subnet-06d67f47275c42f33 / MY_SUBNET_PUBLIC_1a

Created

Wednesday, 16 September 2026 at 13:30:11 GMT+5:30

Deleted

–

Network Routing Tables:-

Route Table Name	Associated Subnets	Destination Target Mapping
Public Route Table	Public Web Subnets 1 & 2	0.0.0.0/0 → Internet Gateway (IGW)
Private App Route Table	Private App Subnets 1 & 2	0.0.0.0/0 → NAT Gateway
Private DB Route Table	Private Database Subnets 1 & 2	No external default route (Isolated internally)

Network & Instance Security Profiles:-

Security Group	Direction	Protocol	Port	Source
Public ALB SG	Inbound	TCP	80	0.0.0.0/0
Public ALB SG	Inbound	TCP	443	0.0.0.0/0
Public ALB SG	Outbound	All	All	Web SG
Web SG	Inbound	TCP	80	Public ALB SG
Web SG	Inbound	TCP	22	Your IP (only if using SSH)
Web SG	Outbound	TCP	5000	Internal ALB SG & all traffic 0.0.0.0/24
Internal ALB SG	Inbound	TCP	5000	Web SG
Internal ALB SG	Outbound	TCP	5000	Application SG
Application SG	Inbound	TCP	5000	Internal ALB SG
Application SG	Inbound	TCP	22	Bastion SG (if using bastion)
Application SG	Outbound	TCP	3306	RDS SG
RDS SG	Inbound	TCP	3306	Application SG
RDS SG	Outbound	All	All	0.0.0.0/0

Security Group snap

Security Groups (7) [Info](#) [Actions](#) [Export security groups to CSV](#) [Create security group](#)

☐	Name	Security group ID	Security group name	VPC ID
☐	INTERNAL-LB-SG	sg-0e0d9bc50e792ccc7	INTERNAL-LB-SG	vpc-081c241c4e4b95465
☐	MY_PROJECT_Applic...	sg-024e0ff4b1415cd24	Application_Tier_SG	vpc-081c241c4e4b95465
☐	MY_PROJECT_IFLB_SG	sg-0417dd926b27c71db	MY_PROJECT_IFLB_SG	vpc-081c241c4e4b95465
☐	MY-PROJECT_WebTi...	sg-05821d0a64d895d6c	MY-PROJECT_WebTier_SG	vpc-081c241c4e4b95465
☐	MY-PROJECT-DB-SG	sg-0d574f5c4e6553bcf	my_bd_sg	vpc-081c241c4e4b95465

Select a security group